

Multi-Agent LLM Evaluation of Counterfactual Explanations for Income Prediction

AFRA Lama, MORTADA Daniel, RIABICHEFF Ivan, DESOIL Théo

Université Libre de Bruxelles

INFO-H512 Project

{lama.afra, daniel.mortada, ivan.riabicheff, theo.desoil}@ulb.be

Abstract

This project studies [selected AI method] in the context of [seminar topic]. We formulate the research question: [insert research question]. We review the relevant literature, implement the method, and evaluate it on [dataset/task]. Our results show that [main result], while also highlighting [main limitation].

Introduction and Problem Formulation

This project connects two seminar themes: **Counterfactuals for Explainability and Fairness in Machine Learning** and **Orchestrate, Secure & Scale Enterprise GenAI Architecture**. It is related to the first seminar because we study counterfactual explanations for a binary income prediction model. Given a person whose income is predicted to be below \$50K, we use DiCE to generate counterfactual examples that would change the model prediction to above \$50K. The central question is not only whether these counterfactuals flip the model output, but whether they are good explanations. In the context of explainable AI, a useful counterfactual should be close to the original input, plausible, actionable, and sparse, meaning that it should change only a small number of features.

The project is also related to **Orchestrate, Secure & Scale Enterprise GenAI Architecture** because we implement a multi-agent LLM pipeline for evaluating these explanations. The seminar discussed generative AI workflows, orchestration of AI services, hallucination control, auditability, human validation, and trustworthy LLM systems. Our pipeline follows this perspective by using several LLM-based roles, including a Prosecutor, Defense agent, Expert Witness, Judge, and Single Evaluator baseline, to assess counterfactual quality in a structured and auditable way.

The main research question addressed in this report is:

Do LLM-based evaluators provide added value over deterministic metrics for assessing counterfactual explanation quality, and does adversarial multi-agent debate improve over a single-LLM evaluator?

This question is relevant because counterfactual generation methods such as DiCE typically select counterfactuals using deterministic criteria. These include **validity**, which checks whether the generated counterfactual flips the model prediction; **proximity**, which favors small changes from the original input; **sparsity**, which favors changing few features; and **feasibility**, which can restrict which features are allowed to change. These metrics are necessary, but they do not fully capture whether a proposed change is realistic, fair, causally coherent, or convincing from a human perspective. We therefore test whether LLM-based evaluators provide additional useful assessment of the counterfactuals generated by DiCE. Since no external ground-truth labels exist for the quality of these counterfactual explanations, we use a deterministic metrics-only evaluator as a rule-based reference standard. This reference encodes our project-specific human judgment through explicit heuristics and thresholds over validity, sparsity, proximity, confidence, and plausibility checks. We then compare a single-LLM evaluator and a multi-agent adversarial debate system against this reference.

We evaluate this question on the OpenML Adult Income dataset, where the task is to predict whether a person's income is above or below \$50K.

Related Work

Taxonomy of the Research Area

Machine learning is increasingly used in high-impact domains such as healthcare, finance, and banking. This makes explainability important, because many useful models are difficult to interpret directly. The research area can be divided into three relevant families of methods:

- **Intrinsically interpretable models:** methods such as linear models or decision trees are easier to inspect because their decision process is relatively transparent. However, they may be less flexible than more complex models.
- **Post-hoc counterfactual explanations:** these methods explain a model decision by asking what minimal changes to the input would have led to a different prediction. They are useful because they can explain individual decisions without requiring access to the internal

structure of the model.

- **Counterfactual quality evaluation:** generated counterfactuals are usually assessed with metrics such as validity, proximity, sparsity, diversity, and feasibility. These metrics are useful but incomplete, because they do not always capture whether a recommendation is realistic, fair, or actionable in the real world.

This taxonomy positions our project in the third family. We do not only generate counterfactual explanations; we study how their quality should be evaluated. In particular, we compare deterministic quality metrics with an LLM-based multi-agent audit that reasons about the same counterfactuals from several perspectives.

Seminal and Recent Works

Counterfactual explanations were introduced as a way to explain automated decisions without opening the black box of the underlying model (Wachter, Mittelstadt, and Russell 2017). Instead of describing all internal model mechanisms, a counterfactual explanation answers a local question: what would need to change in this input for the model to produce a different outcome? For a classification task, this means showing how an unfavorable prediction could be changed into a favorable one.

Later work extended this idea by emphasizing diversity and actionability, rather than validity alone. DiCE generates multiple diverse counterfactual alternatives for a given instance, giving users several possible paths toward the desired outcome instead of a single potentially unrealistic modification (Mothilal, Sharma, and Tan 2020). This is the counterfactual generation framework used in our project.

However, DiCE-style evaluation still relies heavily on numerical criteria. Validity checks whether the model prediction flips; proximity measures how close the counterfactual is to the original instance; sparsity measures how many features change; diversity measures variation among generated alternatives; and feasibility constrains which features may be modified. These criteria are necessary, but they can miss important semantic failures. For example, a counterfactual may be close in feature space while still proposing an impossible change, such as decreasing a person’s age from 30 to 25. Similarly, numerical metrics may not fully capture fairness, realism, or causal coherence.

Our project addresses this gap with a multi-agent collaborative audit of counterfactual explanations. Because there is no external ground truth for these quality labels, our deterministic metrics-only evaluator acts as the reference standard: it encodes our human judgment through explicit rules over metrics and issue-specific plausibility checks. The LLM evaluators are then assessed by how well they reproduce, justify, or meaningfully challenge this rule-based assessment. To make the audit operational, we define a small issue taxonomy covering implausible time-dependent changes, extreme working hours, unactionable capital shifts, too many simultaneous changes, and fragile counterfactuals;

the next section describes how these labels are used in the evaluation pipeline.

Method

The original predictive task is binary classification on the Adult Income dataset. Each instance describes a person using demographic, educational, occupational, financial, and working-time attributes, such as age, education, occupation, capital gain and loss, hours worked per week, and native country. The unfavorable class corresponds to a predicted annual income of at most \$50K, while the favorable class corresponds to a predicted annual income above \$50K. We train a Logistic Regression model inside an `sklearn` pipeline. This model is used for counterfactual generation because DiCE requires reliable class-probability outputs that can guide its optimization procedure.

Before generating counterfactuals with DiCE, we define a feature policy that freezes attributes that should not be modified: `fnlwgt`, marital status, relationship, race, sex, and native country. Counterfactuals are therefore not allowed to assign different values to these features, since doing so would imply unfair or unrealistic demands on the individual.

After training the predictive model and applying this feature policy, we generate counterfactual explanations with DiCE. For each individual with an unfavorable prediction, DiCE produces multiple candidate counterfactuals that aim to move the prediction to the favorable class. We keep the DiCE generation parameters at their default values and use the feature policy only to restrict which attributes may change.

For each generated counterfactual, we compute a first set of standard numerical quality metrics:

- **Validity:** whether the counterfactual reaches the favorable class.
- **Proximity:** how close the counterfactual remains to the original instance.
- **Sparsity:** how many features are changed.
- **Diversity:** how different the generated counterfactuals are from each other.
- **Counterfactual confidence:** the model probability assigned to the favorable class for the counterfactual.

These metrics are useful for ranking and filtering counterfactuals, and they are close to the criteria commonly used in DiCE-style generation. However, they remain purely numerical. They do not directly represent whether a counterfactual is plausible, actionable, or convincing from a human perspective. We therefore add an issue taxonomy that provides a second evaluation layer. The taxonomy defines a fixed vocabulary of possible counterfactual failures:

- `implausible_time_dependent_change`: age decreases, education decreases, education increases without sufficient age increase, or age/education becomes non-integer.

- `extreme_working_hours`: weekly working hours become dangerously high, implausibly low, or change by an unrealistic amount.
- `unactionable_capital_shift`: capital gain or capital loss changes in a financially unrealistic way.
- `too_many_changes`: too many actionable features are changed at the same time.
- `fragile_counterfactual`: the counterfactual only barely reaches the favorable class, with confidence close to the decision threshold.

Each label corresponds to a specific failure mode in the generated counterfactual. This taxonomy gives all evaluation strategies the same vocabulary: the metrics-only evaluator, the single-LLM evaluator, and the multi-agent LLM evaluator all reason over the same issue labels. It also constrains the LLM-based evaluators by discouraging free-form criticism that is not grounded in the predefined evaluation framework.

We operationalize the taxonomy through deterministic heuristic rules. These rules encode human-designed plausibility checks over the original instance, the counterfactual, and the numerical metrics. For example, they flag decreases in age, unrealistic changes in education or working hours, large unexplained capital shifts, excessive simultaneous modifications, and low-margin favorable predictions. The heuristic module returns the set of issue labels detected for each counterfactual.

Finally, we define a metrics-only deterministic evaluator. This evaluator does not use an LLM. It receives as input the generated counterfactuals, their standard numerical metrics, and the heuristic issue flags derived from the issue taxonomy. It outputs two judgments for each counterfactual:

- **Issue severity**: `low`, `medium`, or `high`, depending on the number and type of detected issues.
- **Overall assessment**: `fair`, `ambiguous`, or `unfair`, summarizing whether the counterfactual is acceptable as an explanation.

This deterministic evaluator acts as the rule-based reference against which the LLM-based evaluation strategies are compared.

Single-LLM evaluator. The second evaluation strategy is a single LLM baseline. It receives the same information as the metrics-only evaluator, but formatted as a structured prompt. The prompt contains the original individual profile, the generated counterfactuals, the feature changes, the numerical metrics, the heuristic issue evidence, and the taxonomy instructions. The model then produces one structured verdict for each counterfactual, including an overall label, such as `fair`, `ambiguous`, or `unfair`, and a short justification. This baseline tests whether LLM reasoning alone adds value over deterministic metrics before introducing multi-agent debate.

Multi-agent adversarial debate. The central model

of the project is an LLM-based multi-agent debate system implemented with the Python framework AutoGen. This system operationalizes the main research question by testing whether adversarial role-based discussion improves counterfactual evaluation beyond a single LLM. It uses four specialized agents:

- **Prosecutor**: attacks the counterfactual and identifies possible flaws.
- **Defense attorney**: defends the counterfactual and argues for its validity, plausibility, and usefulness.
- **Expert witness**: summarizes the numerical metrics and the heuristic evidence derived from the issue taxonomy.
- **Judge**: synthesizes the previous arguments into a final structured verdict.

The judge produces the final assessment for each counterfactual in the same format as the other evaluators, making the three strategies directly comparable.

All LLM-based components, including the single-LLM evaluator and the four debate agents, are controlled through structured prompts. These prompts contain explicit rules:

- use only issue labels from the predefined taxonomy;
- do not invent new categories of counterfactual criticism;
- rely on the heuristic flags when discussing plausibility and validity;
- discuss flip robustness using `cfl_confidence`, where a counterfactual is considered fragile when its confidence is close to the 0.5 decision threshold.

These constraints make the LLM outputs auditable and keep all evaluators aligned with the same evaluation framework.

Experimental Setup

Dataset

We evaluate the method on [dataset name]. The dataset contains [brief description: number of samples, features, classes, task, etc.].

Baselines

We compare [selected method] against the following baselines:

- **Baseline 1**: [description];
- **Baseline 2**: [description].

Metrics

We use the following metrics:

- **Metric 1**: [why appropriate];
- **Metric 2**: [why appropriate];
- **Metric 3**: [why appropriate].

Method	Metric 1	Metric 2	Metric 3
Baseline 1	–	–	–
Baseline 2	–	–	–
Selected method	–	–	–

Table 1: Experimental results comparing the selected method with baselines.

Implementation Details

The implementation is written in Python. The code is available at:

<https://github.com/your-username/your-repository>

The repository includes instructions to reproduce the experiments, including dependencies, dataset preparation, and commands used to generate the results.

Results and Analysis

Table 1 summarizes the main experimental results.

The results show that [main observation]. Compared with [baseline], the selected method [performs better/worse] in terms of [metric]. This suggests that [analysis].

However, the method also presents limitations. First, [limitation 1]. Second, [limitation 2]. These limitations are important because they affect [robustness, scalability, interpretability, feasibility, etc.].

Figure ?? illustrates [what the figure shows].

Discussion

The experiments indicate that [answer to research question]. The main strength of [selected method] is [strength]. This makes it suitable for [setting].

Nevertheless, the method assumes that [assumption]. In practice, this assumption may not hold when [case]. Another limitation is [another limitation].

Possible improvements include [future improvement], [extension], or combining the method with [other approach].

Conclusion

This report studied [selected method] for [task/problem]. We reviewed the relevant literature, implemented the method, and evaluated it experimentally. Our findings suggest that [main conclusion]. Future work could investigate [future direction].

References

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